

COMPUTER NETWORKS: MASTER TEXTBOOK

1. Introduction & Network Models

OSI Model (7 Layers)

The Open Systems Interconnection (OSI) model standardizes network functions into 7 layers:

- **Physical**: Transmits raw bit streams (cables, hubs).
- **Data Link**: Frame delivery, MAC addressing (switches).
- **Network**: Packet routing, IP addressing (routers).
- **Transport**: End-to-end communication, error correction (TCP/UDP).
- **Session, Presentation, Application**: Data formatting and user interface.

2. TCP/IP Protocol Suite

Transport Layer Protocols

TCP (Transmission Control Protocol) Connection-oriented, reliable, guarantees packet delivery.

UDP (User Datagram Protocol) Connectionless, fast, no error recovery. Ideal for real-time streaming.

3. IP Addressing and Routing

Addressing

IPv4: 32-bit address space, dotted-decimal notation.

IPv6: 128-bit address space, alphanumeric notation to handle the explosion of internet-connected devices.

4. Network Security

Foundations

Modern networking relies on **Encryption**(SSL/TLS for secure communication) and **Firewalls**to monitor and filter traffic. Understanding threats like DoS (Denial of Service) and Spoofing is critical for network administrators.